

Detecting Abnormal Markets

Early Warning Systems with Deep Anomaly Detection

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Why this matters

Markets crash often enough to matter – 1929, 2000, 2008, 2020 ...

Missing the 10 worst days vs. buy-and-hold S&P 500 (1927–2015):

Strategy	Growth of \$1
Total cumulative return	\$115.40
Miss the 10 worst days	\$362.01
Miss the 10 best days	\$38.28

Tail events cluster

Frequency and cross-asset correlations of simultaneous heavy losses keep rising. Diversification fails exactly when you need it most.

The problem: risk-on → risk-off

Risk-on

Investors are confident. They bid up equities, credit, EM debt, commodities.

Volatility is low. Spreads compress. Capital flows toward growth.

Risk-off

Investors are scared. They flee to cash, short-term govies, gold, CHF, JPY.

Volatility spikes. Correlations surge. This is systemic risk.

- We do not need to *forecast* the crash – we need to *nowcast* the transition.
- Goal: detect the green-to-red turn as early as possible.

Solution: Early Warning Systems

- Treat **normal markets** as the inlier distribution and **stress regimes** as anomalies.
- Train novelty detectors on the normal samples only; raise a defensive flag when the anomaly score crosses a threshold.
- Why this framing beats outright crash prediction:
 - No need to label every crisis type ex ante.
 - The model scores the latest weekly observation in real time.
 - It generalises – a 2008-style stress and a 2020-style stress both look “not normal”.

Deployment

A defensive flag rotates the portfolio into cash, short-term govies, gold, or other low-beta assets. Typical user: CIO, risk officer, wealth manager.

Bloomberg weekly panel

- Key equity & bond indices (IG/HY, inflation-linked, mortgages)
- Short / medium / long interest rates
- Major FX, commodities, VIX
- Economic Surprise, Baltic Dry
- Binary label: *normal* vs. *abnormal*

Same model, two business cases

- **Risk management** – reduce drawdown
(*this work*)
- Quant strategy – generate alpha

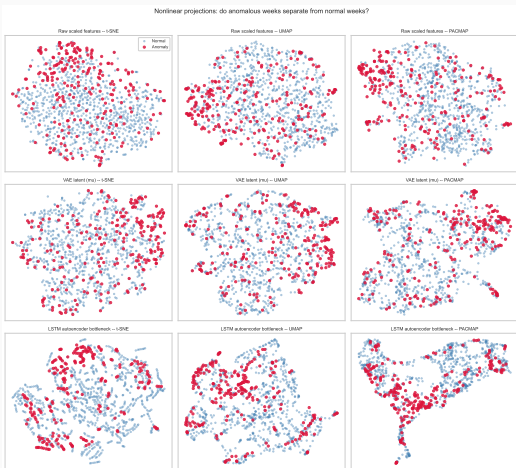
That choice drives data treatment, threshold tuning, and the evaluation metric we report.

Three deep novelty detectors

All three are trained on **normal observations only**; high score $\Rightarrow$ defensive flag.

- **GAN** – generator learns the normal manifold; two scoring heads
 - Discriminator realism: $s(x) = 1 - D_{\text{prob}}(x)$
 - Reconstruction in latent space: $s(x) = \min_z \|x - G(z)\|^2$
- **VAE** – reconstruction error of a Gaussian-regularised latent space; KL-regularised encoder gives a smoother score.
- **LSTM autoencoder** – **bidirectional 2-layer** encoder over a 12-week rolling window; scored by an **ensemble of last-step reconstruction MSE and Mahalanobis distance** in the bottleneck latent.

Visual evidence: do anomalous weeks actually separate?



Raw features

Partial separation only.

VAE latent

Anomalies stay near the normal manifold.

LSTM bottleneck

Clear t-SNE/UMAP cluster;
PACMAP places anomalies on the manifold perimeter.

Threshold choice and evaluation

Threshold tuned on cross-validation

Maximise F_β with $\beta = 1.5$, subject to a precision floor of $\min(0.7, \text{base rate} + 0.25)$.

Headline metrics for the risk officer:

- **Recall** – fraction of real crises the detector would catch.
- **PR-AUC** – threshold-free summary that stays honest under heavy class imbalance.
- **False-negative rate** – a missed crisis is the expensive failure mode.

Precision, F1, ROC-AUC are reported for completeness but are not the primary lens: a defensive false alarm costs basis points; a missed crash costs a drawdown.

Out-of-sample detector results

Best recall
86.6%

LSTM autoencoder

Lowest missed-crisis rate
13.4%

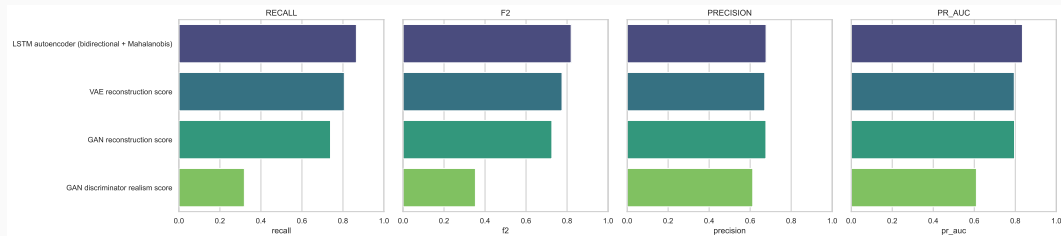
LSTM autoencoder

Best PR-AUC
83.5%

LSTM autoencoder

Best F2
0.820

LSTM autoencoder

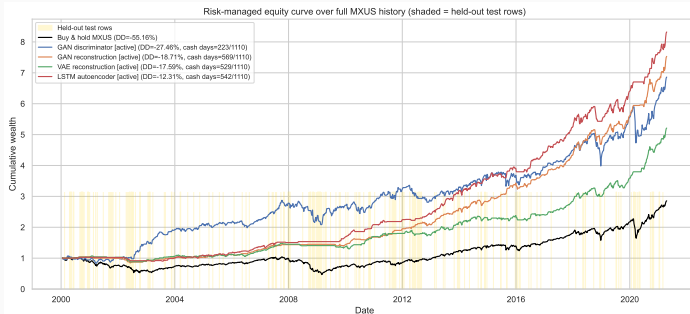


Model	Recall	FNR	PR-AUC	Precision
LSTM autoencoder	86.6%	13.4%	83.5%	67.8%
VAE reconstruction	80.7%	19.3%	79.5%	67.1%
GAN reconstruction	74.0%	26.1%	79.7%	67.7%
GAN discriminator	31.9%	68.1%	61.0%	61.3%

Risk-management read

LSTM sweeps every metric: highest recall, lowest missed-crisis rate, best PR-AUC, and best precision – the bidirectional encoder + Mahalanobis latent score is doing the work.

Portfolio impact: drawdown reduction



Backtest summary

Strategy	Wealth	Max DD
Buy & hold MXUS	2.86	-55.2%
GAN discriminator	6.86	-27.5%
GAN reconstruction	7.53	-18.7%
VAE reconstruction	5.21	-17.6%
LSTM autoencoder	8.32	-12.3%

- **LSTM dominates:** best wealth (8.32) *and* best drawdown (-12.3%).
- Beats buy & hold on both axes simultaneously.